

Life-history trait and functional diversity patterns of ground beetles and spiders along a coastal heathland successional gradient

Jens Schirmel^{a,b,*}, Irmgard Blindow^a, Sascha Buchholz^c

^aBiological Station of Hiddensee, University of Greifswald, Biologenweg 15, 18565 Isle of Hiddensee, Germany

^bInstitute for Environmental Sciences, University of Koblenz-Landau, Fortstr. 7, 76829 Landau, Germany

^cDepartment of Ecology, TU Berlin, Rothenburgstr. 12, 12165 Berlin, Germany

Received 12 April 2012; accepted 28 August 2012

Abstract

We investigated patterns in species richness and diversity, life-history traits and functional diversity of carabid beetles and spiders along a coastal heathland successional gradient. We sampled in five successional stages on the island of Hiddensee, Germany. Species richness of carabids and spiders and Simpson diversity for spiders did not differ among stages. Functional dispersion (FDis), a measure of functional richness, was lowest in the youngest stage for both carabids and spiders. Older successional stages represent more heterogeneous habitats and offer a broader range of niches, which might enhance functional dispersion. Functional evenness (FEve) differed among successional stages for spiders, indicating an uneven distribution of species abundances in the functional trait space. Functional divergence (FDiv) did not differ among stages for either taxon. Shifts in life-history traits were noted in both arthropod groups, but not always in the same direction: Body size of carabid beetles increased towards older successional stages, but decreased for spiders. The number of flightless carabid species increased towards older habitats. Each successional stage seemingly favors species with distinct life-history traits (species sorting). We conclude that a trait-based approach including the use of functional diversity measurements (FDis) could contribute valuable information for biodiversity conservation.

Zusammenfassung

Ziel der vorliegenden Studie ist die Analyse von Diversitätsmustern (alpha-Diversität, funktionale Diversität) und dem Vorkommen biologischer Merkmale von Laufkäfern und Spinnen entlang eines Sukzessionsgradienten in einer Küstenheide auf der Insel Hiddensee (Deutschland). Beprobt wurden fünf Stadien mit jeweils drei Wiederholungen. Sowohl für die Artenzahlen als auch für die Simpson-Diversität der Spinnen ließen sich keine Unterschiede zwischen den Sukzessionsstadien feststellen. Die funktionale Diversität (functional dispersion, FDis) beider Taxa war in den jüngsten Stadien am niedrigsten, wohingegen die älteren Stadien heterogenere Habitate sind. Daraus ergibt sich ein breiteres Nischenangebot, das die funktionale Diversität möglicherweise erhöht. Die Unterschiede in der funktionalen Evenness (functional evenness FEve) der Spinnen sind vermutlich auf eine ungleiche Verteilung der Merkmalshäufigkeiten zurückzuführen. Für die funktionale Divergenz (functional divergence, FDiv) ergaben sich keine Unterschiede entlang des Sukzessionsgradienten. In beiden Artengruppen wurden Unterschiede in der Verteilung der biologischen Merkmalen festgestellt: Die Körpergröße von Laufkäfern nahm mit fortschreitender Sukzession zu,

*Corresponding author at: Institute for Environmental Sciences, University of Koblenz-Landau, Fortstr. 7, 76829 Landau, Germany.

Tel.: +49 6341 28031535; fax: +49 6341 28031185.

E-mail address: schirmel@uni-landau.de (J. Schirmel).

während Gegenteiliges für die Spinnen beobachtet wurde. Weiterhin stieg die Anzahl flugunfähiger Laufkäfer mit dem Sukzessionsalter. Jedes Sukzessionsstadium scheint das Vorkommen von Arten mit bestimmten biologischen Merkmalsausprägungen zu begünstigen (species sorting). Sowohl die Analyse von Verteilungsmustern biologischer Merkmale als auch die Interpretation funktionaler Diversitäts-Indizes kann einen wichtigen Beitrag für den Biodiversitätsschutz leisten.

© 2012 Gesellschaft für Ökologie. Published by Elsevier GmbH. All rights reserved.

Keywords: Araneae; Carabidae; Functional dispersion; Simpson diversity; Species richness; Species sorting

Introduction

Succession in open terrestrial ecosystems is a key driver of biodiversity dynamics and can have differential effects on invertebrate diversity (e.g., Balmer & Erhardt 2000; Marini, Fontana, Battisti, & Gaston 2009; Buchholz 2010). In heathlands, without management the depletion of nutrients is inhibited and vegetation undergoes succession towards shrub or tree dominated vegetation (Webb, 1998). In such ecosystems, successional processes often are detrimental to open-habitat specialist or xerothermic species due to e.g., loss of disturbance and dynamic effects (Bonte, Lens, & Maelfait 2006; Schirmel & Buchholz 2011). For analyzing shifts in terrestrial ecosystem conditions the ground-dwelling arthropods carabid beetles (Coleoptera: Carabidae) and spiders (Araneae) are suitable indicator taxa (carabids: e.g., Lövei, Magura, Tothmeresz, & Ködöböcz 2006; Lin, James, & Dolman 2007; spiders: e.g., Huber, Schulze, & Baumgarten 2007; Horvath, Magura, Szinetar, & Tothmeresz 2009).

Within the framework of biodiversity and conservation studies, species richness is usually the simplest and most intuitive measure for diversity (Magurran 2004). Several diversity indices have been developed to incorporate the relative abundance of species (or the evenness): of these the Simpson index D is intuitively meaningful and provides good diversity estimates (Magurran 2004). Both species richness and (Simpson) diversity have often been used in ecological studies to describe assemblage shifts and they have proven useful tools to assess and monitor changes caused by environmental variability (e.g., Blaum, Seymour, Rossmannith, Schwager, & Jeltsch 2009; Negro, Isaia, Palestini, Schoenhofer, & Rolando 2010; Schirmel, Timler, & Buchholz, 2011). Apart from these ‘traditional’ diversity measures, the concept of ‘functional diversity’, which integrates information on life-history traits has grown rapidly in recent years in ecological research (Petchey & Gaston 2006; Violle et al. 2007; Il Choi et al. 2010), since the realization that life-history traits also have an important role to play in diversity (Ribera, Dolédec, Downie, & Foster 2001; Da Silva, Aguiar, Niemelä, Sousa, & Serrano 2007; Vandewalle et al., 2010). Zoological studies have mainly applied the trait concept to freshwater and marine environments (Bremner, Rogers, & Frid 2006; Diaz, Alonso, & Gutierrez 2008), but also to terrestrial arthropods (Keil, Dziöck, & Storch 2008; Lambeets, Vandengehuchte, Maelfait, & Bonte 2008; Lambeets, Vandengehuchte, Maelfait, & Bonte 2009). Functional diversity concepts can provide a useful approach to

integrate biodiversity research into the broader context of ecosystem processes and functioning (Loreau 2010), e.g., for productivity (Tilman et al. 1997) or resilience (Folke et al. 2004). Several indices were developed analogously to traditional alpha diversity indices to measure functional diversity or to display functional composition (Petchey & Gaston 2006; Villéger, Mason, & Mouillot 2008; Laliberté & Legendre 2010). In the present study, three indices were used to explore the different facets of functional diversity: functional evenness (FEve) and functional divergence (FDiv) (Villéger et al. 2008) as well as functional dispersion (FDis) (Laliberté & Legendre 2010). Although it has an extensive theoretical and mathematical background (Schleuter, Daufresne, Massol, & Argillier 2010), functional diversity, especially from multiple traits, has rarely been incorporated into ecological studies to date, and little is known in particular about shifts of functional diversity along environmental gradients (Gerisch et al. 2011). However, life-history traits of carabids and spiders have already been successfully integrated into investigations of assemblage shifts (e.g., Lambeets et al. 2008; Lambeets et al. 2009; Il Choi et al. 2010; Vandewalle et al. 2010).

In their metacommunity theory, Leibold et al. (2004) argue that the community composition in spatially structured, heterogeneous environments – such as coastal heathlands – could be altered by ‘species sorting’. In species sorting, resource gradients cause differences in species composition because each species becomes a specialist for different local habitat conditions. The concept presumes that local habitat templates can act as filters sorting species according to their functional traits (Leibold et al. 2004). As a consequence, only species with functional traits that match the specific environment can pass these filters (Poff 1997).

In particular, we (i) tested whether species richness and diversity as well as functional diversity increased with age and structurally complexity of successional stage and (ii) hypothesized that succession is reflected by shifts in life-history traits, and (iii) that certain traits are responsible for species sorting along a successional gradient.

Materials and methods

Study area

The southern Baltic Sea island of Hiddensee is about 16 km² and is situated in the National Park ‘Vorpommersche

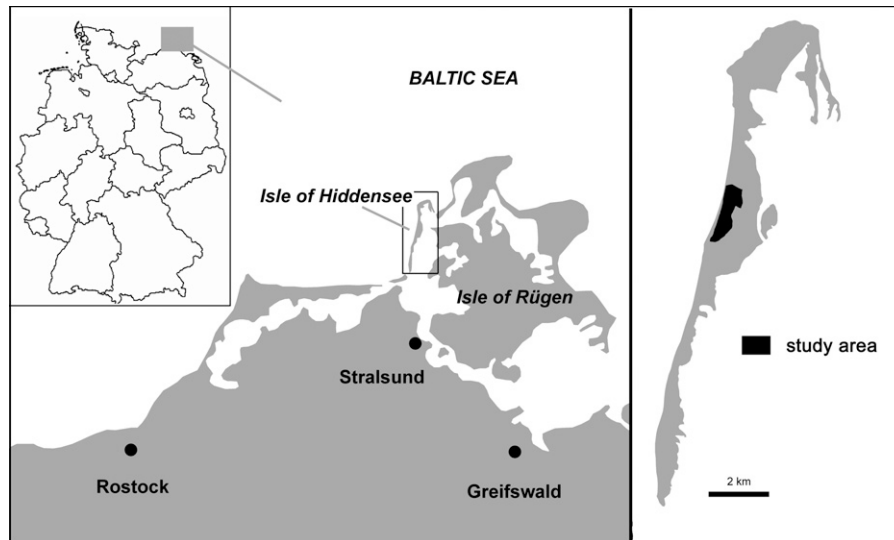


Fig. 1. Position of the Isle of Hiddensee in Germany and of the study area on the island.

Boddenlandschaft' (Western Pomeranian lagoon area) in Northeastern Germany, Mecklenburg-Western Pomerania (Fig. 1). The climate is characterized by an average annual temperature of 7.5 °C and an average annual precipitation of 547 mm (Reinhard 1962). An anthropo-zoogenic coastal heathland is situated in the centre of the island (54°32'N, 13°5'E; 1–5 m a.s.l.; Fig. 1) on acidic, nutrient-poor and sandy soils (Bauer 1972). This area was traditionally used as grazing ground for domestic animals and for fuel and building material. However, traditional land use decreased at the beginning of the 20th century and ceased totally around the Second World War (Umweltministerium Mecklenburg-Vorpommern 2003). Nowadays it is preserved as an open landscape by several management measures such as shrub clearing, mowing, sod cutting and (since 2004) sheep grazing. Five main stages of (coastal) heathland succession can be found (Remke 2003): (i) open acidic (coastal) dunes (grey dunes) dominated by *Corynephorus canescens*, *Carex arenaria* and cryptogams representing the youngest successional stage (G); (ii) dwarf-shrub heath dominated by *Calluna vulgaris* and *Empetrum nigrum*, the main vegetation type within the study area (H); (iii) grassy heath, where the majority of the dwarf-shrubs are displaced by tall graminoids, especially by *C. arenaria* (C); (iv) shrub-heathland dominated by *Betula pendula*, *Betula pubescens* or *Prunus serotina* (S); (v) young birch forest, which has developed due to the lack of management during the last 30–40 years, thus representing the oldest successional stage (B). The five successional stages differed in habitat heterogeneity (expressed as the Simpson-Diversity of estimated cover of 'trees', 'shrubs', 'field layer', '*C. vulgaris*' and 'cryptogams' based on vegetation sampling data from Schirmel, Mantilla-Contreras, Blindow, and Fartmann (2011); one-way ANOVA: $F=53.901$, $P<0.001$). Habitat heterogeneity was lowest in

grey dunes (mean and SE: 0.53 ± 0.02) and increased almost continuously towards the older successional stages dwarf-shrub heath (0.66 ± 0.00), grassy heath (0.58 ± 0.01), heath with shrubs (0.68 ± 0.01), and birch forest (0.69 ± 0.01).

Sampling of carabid beetles and spiders

A total of 15 plots (three replicates per successional stage) were established. Each plot was 500 m² and had a homogeneous vegetation structure typical for the stage in question. Carabid beetles and spiders were sampled using pitfall traps. Pitfall trapping is a standardized method widely used for sampling ground-dwelling arthropods, producing high catch numbers and useful data on activity densities (New 1998). In each plot, four pitfall traps were randomly set level with the soil surface, but at a distance of at least 5 m from each other to minimize potential interference and to the boundaries to minimize edge effects. The traps were 6 cm in diameter, 8 cm deep and filled with a 70% ethylene-glycol solution and a few drops of detergent. A 15 cm × 15 cm transparent plastic roof was placed a few cm above the trap to exclude litter infall and precipitation. Pitfalls were active between 15 May 2008 and 04 May 2009 and were emptied monthly. We sampled year-round to avoid possible biases due to inter-specific differences in seasonal activities. Thus, our dataset delivered information on the activity densities of individual species in a specific habitat (e.g., Baars 1979; Ribera et al. 2001).

Catches were sorted into taxa and stored in 70% alcohol. Carabid beetles were identified to species level according to Müller-Motzfeld (2006), spiders according to Roberts (1987, 1998) and Nentwig et al. (2003). Only adult spiders were included in the analysis. Nomenclature follows

Table 1. Life-history traits of carabid beetles and spiders based on literature data.

Trait	Category
<i>Carabids</i>	
Mean body size ^{a,b}	Very small (<5 mm) Small (5–9 mm) Medium (9–15 mm) Large (>15 mm)
Ability of active flight ^c	Flightless Able to fly
Feeding ^a	Collembola specialist Generalist predator Mixed feeder Phytophagous
<i>Spiders</i>	
Mean body size of females ^{d,e}	Very small (<3.1 mm) Small (3.1–4.5 mm) Medium (4.6–8.4 mm) Large (>8.5 mm)
Ballooning ^{i,f}	Very common Less common Uncommon
Hunting ^{d,g,h}	Active hunting Web Ambush

^aMüller-Motzfeld (2006).^bRibera, Foster, Downie, McCracken, and Abernethy (1999).^cTurin (2000).^dNentwig et al. (2003).^eRuzicka (1985).^fBell et al. (2005).^gRoberts (1987).^hRoberts (1998).

ⁱSpider species which are unable to balloon as adults are often categorized as “non-ballooning” species. However, some of these species are able to balloon as juveniles. We therefore choose less strict categories for ballooning: “very common”: both adults and juveniles regularly balloon, “less common”: both adults and juveniles irregularly balloon, “uncommon”: non-ballooning adults with sometimes ballooning juveniles.

Müller-Motzfeld (2006) (carabids) and Platnick (2012) (spiders).

Life-history traits

Carabid beetles and spiders were assigned to life-history trait categories with the help of literature data and using morpho-physiological and phenological features which potentially affect the occurrence patterns of these species (Ribera et al. 2001; Bonte et al. 2006; Lambeets et al. 2009) (Table 1; Appendix A: Table 1). The following traits were analyzed: ‘body size’, ‘dispersal’ and ‘feeding preference (carabids)/hunting mode (spiders)’. For body size, metric data (mm) were given in addition to body size categories.

Data analysis

Diversity

For each plot, data from the four pitfall traps were pooled, and species and individual catch numbers were used for statistical analyses. Species richness was expressed as number of species per plot and we chose the Simpson index as a diversity measure (1-D).

We used three functional diversity measurements. FDis expresses changes in functional dispersion, FEve for functional evenness and FDiv for functional divergence. FDis is a measure of functional richness, which considers species relative abundances by estimating their dispersion in a multidimensional trait space (Laliberté & Legendre 2010). In the following, we interpret functional dispersion as a measure for functional diversity per se. Life-history trait data (for dispersal, feeding preference/hunting mode) were coded categorically for calculating functional diversity indices except body size. All used indices are distance-based and can be computed for categorical variables (Laliberté & Legendre 2010; Laliberté & Shipley 2010; Schleuter et al. 2010). We computed FDis (Laliberté & Legendre 2010) and FEve and FDiv (Villéger et al. 2008) using the R language package FD (Laliberté & Shipley 2010). Differences in species richness, Simpson diversity and functional diversity indices among the successional stages were tested with generalised linear models (GLMs) using Poisson models for count data and Gaussian error structures and identity link function for indices.

Life-history traits

To detect possible differences in life-history trait composition among successional stages, we applied GLMs with the number of species of a trait category per plot as the dependent variable (e.g., number of very small spider species per plot). To compensate for overdispersion, the standard errors were corrected using a quasi-Poisson model (Zuur, Ieno, Walker, Saveliev, & Smith 2009). In addition, mean body size of carabid beetles and spiders were compared among the five successional stages using one-way ANOVA.

The free software package R 2.11.1 (R Development Core Team 2010) was used for all analyses.

Results

Diversity

In total, 3.646 carabid and 9.934 spider individuals were trapped (Appendix A: Table 1). Species richness did not differ among the five successional stages for carabid beetles or spiders (Appendix A: Table 2). In contrast, Simpson diversity differed among the stages for carabids (Fig. 2), while no differences were observed for spiders (Appendix A: Table 2). FDis of both carabid beetles and spiders differed significantly among the successional stages and was lowest in grey dunes (Fig. 2), while no differences were observed in

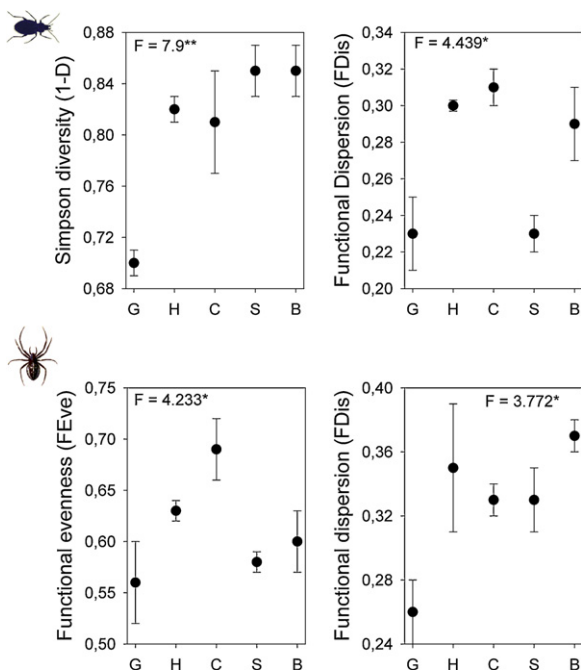


Fig. 2. Diversity measures (mean values + SE) of carabid beetles (Simpson diversity, functional dispersion) and spiders (functional evenness, functional dispersion) showing significant differences among five heathland successional stages on the Baltic island of Hiddensee, Germany. Significance of differences was tested using Gaussian GLM with identity link. G: grey dunes, H: dwarf-shrub heath, C: grassy heath, S: heath with shrubs and B: birch forest.

FDiv (Appendix A: Table 2). For spiders, FEve differed significantly among stages and was also lowest in grey dunes (Fig. 2). No differences in FEve were found for carabids (Appendix A: Table 2).

Life-history traits

For the tested life-history traits, frequencies of trait categories within plots differed significantly among the five stages (Fig. 3). Carabid beetle body size categories differed significantly and mean number of very small species was highest in grey dunes while mean number of large species was highest in the two oldest stages. Accordingly, mean body size differed significantly among stages and was lowest in grey dunes. Dispersal ability differed significantly among stages, with the lowest number of flightless species occurring in grey dunes. Feeding preferences of carabid beetles differed in two categories: Collembola predators were most species rich in the oldest stages heath with shrubs and birch forest, while the highest number of phytophagous species occurred in grey dunes.

In contrast to carabid beetles, most large and medium-sized spider species were found in grey dunes and the mean body size also decreased from young to older habitats. Spiders with no ballooning ability were most species rich in grey dunes, followed by dwarf-shrub heath. Species richness of actively hunting spiders was highest in grey dunes.

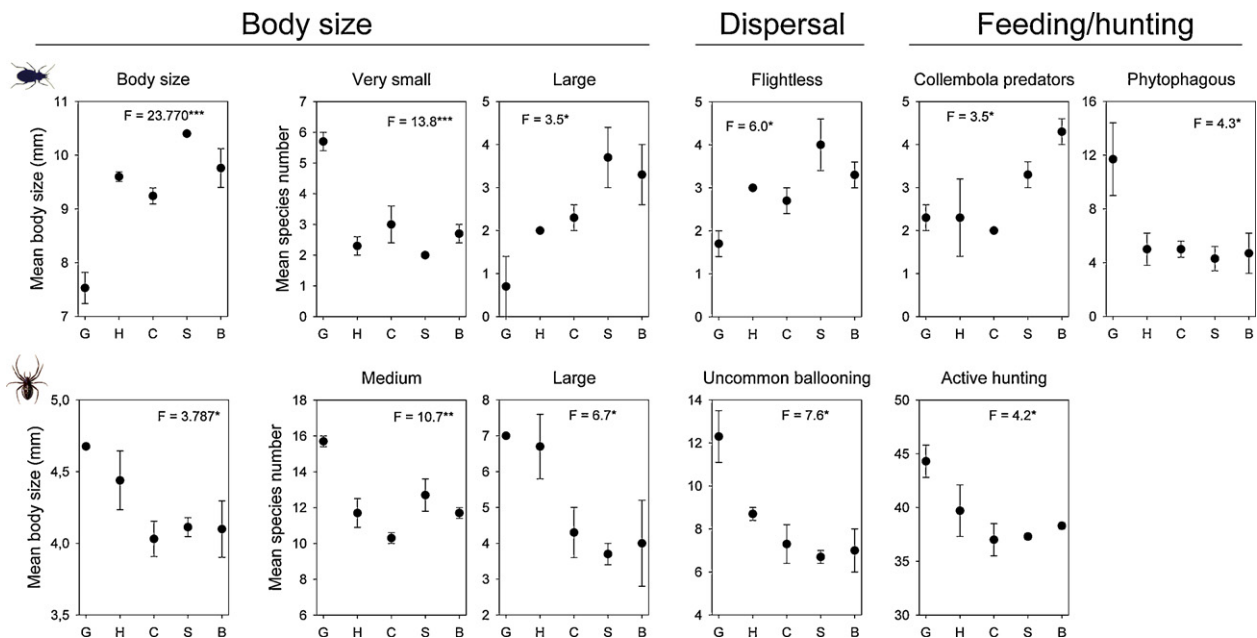


Fig. 3. Mean body size (mean + SE) and species number (mean values + SE) of life-history trait categories of carabid beetles and spiders along a heathland successional gradient. Differences were tested using GLM. Only variables significantly different among stages are shown. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. Non-significant life history-trait variables – for carabids: body size (small, medium), dispersal (able to fly), feeding preference (general predators, mixed feeders); for spiders: body size (very small, small), dispersal (ballooning ability), hunting mode (web hunter, ambush hunter). G: grey dunes, H: dwarf-shrub heath, C: grassy heath, S: heath with shrubs, and B: birch forest.

Discussion

Diversity

With respect to the alpha-diversity measurements species richness and Simpson diversity, differences among the five successional stages were only found in Simpson diversity for carabid beetles (lowest in grey dunes). For spiders, no changes in species richness and Simpson diversity were detected among successional stages, which is in accordance with the findings of previous studies (e.g., Buddle, Spence, & Langor 2000; Gajdos & Toft 2000; Mallis & Hurd 2005). In contrast, Cristofoli, Mahy, Kekenbosch, and Lambeets (2010) showed that spider diversity increased towards older successional stages of wet heathlands. However, diversity is a matter of scale, and mechanisms such as species sorting might be responsible for structuring carabid beetle and spider assemblages or causing assemblage shifts (Lambeets et al. 2009). In species sorting, species distribution is linked to local conditions (niche separation) and is independent of unrelated purely spatial effects (Leibold et al. 2004). Community composition might therefore change at each successional stage by replacing one specialist by another (including shifts in life-history traits) without affecting the overall diversity of the community at the given stage (Buchholz 2010).

For both taxa, FDis was significantly lowest at the grey dune stage. Low functional diversity might be related to a lower number of available niches. Moreover, it might indicate a reduced productivity, since some resources potentially available to the community are unused or not available (Petchey, Evans, Fishburn, & Gaston 2007; Villéger et al. 2008). However, the dominance of a few specialized species, which have become accustomed to (or have adapted to) disturbance effects and extreme microclimate conditions in these young, open, and dynamic habitats might also be responsible for lower functional diversity. In contrast, the older successional stages represent more heterogeneous and complex habitats than the grey dunes and therefore offer broader niche ranges. This could enhance functional diversity and might lead to more (functional) complex species assemblages. High values of FEve indicate low habitat dynamics and balanced niche occupancy (Schleuter et al. 2010). Villéger et al. (2008) noted that both functional and species evenness are closely related. Thus, FEve should be highest in older successional stages with reduced habitat dynamics and lowest in dynamic grey dunes, which corresponded to our results for spiders in this study.

Most obviously, habitat heterogeneity facilitates niche differentiation in coastal heathlands (Schultz & Finch 1996); this is reflected by a higher functional diversity (FDis) of carabid beetles and spiders in later and structurally more complex successional stages. Our findings therefore emphasize that species richness (and to some extent Simpson diversity) have limited explanatory power for analyzing shifts along environmental gradients, as also found e.g., by Brose (2003), Mallis and Hurd (2005) and Buchholz (2010).

Life-history traits

Changes in functional diversity (FDis) were linked to community-wide character shifts of both carabid beetles and spiders. Habitat conditions constrained species composition (Schirmel & Buchholz 2011) but varied among functional groups. Species sorting occurred, allowing species with distinct life-history traits to occur under the given conditions of a certain successional stage.

Carabid beetles were smaller in grey dunes and large-bodied in most vegetated successional stages. This corroborates the stress hypothesis according to which species often have a smaller body size in disturbed and dynamic habitats (Gray 1989). As an adaption to unstable conditions, species in more disturbed habitats often have a high turnover rate and therefore a smaller body size, while species in undisturbed habitats often have a low turnover rate and a larger size. For carabid beetles, support for this hypothesis had already been given by Blake, Foster, Eyre, and Luff (1994), Ribera et al. (2001) and Niemelä and Kotze (2009) who found that body size was smaller in disturbed habitats.

Opposite patterns were observed in spiders, where most large and medium-sized species occurred in grey dunes, implying that body size patterns seem to be constrained by morphology rather than habitat dynamics. Remmert (1981) and Entling, Schmidt-Entling, Bacher, Brandl, and Nentwig (2010) suggested that dry and hot habitat conditions favor large-bodied species because of the higher desiccation resistance which results from their more compact and waterproof cuticle and their lower surface area: volume ratio. Moreover, large spider species with a burrowing life style often occur in dune habitats which benefit from high sand dynamics that avoid an increase in soil hardness and vegetation encroachment (Bonte et al. 2006). However, as only body size values of adults were used for analyses it is possible that patterns may change substantially after including all instars.

In temporary habitats, many carabid species are highly mobile and able to fly as an adaption to fluctuations in habitat availability (Greenslade & Southwood 1962; Bonte, Baert, & Maelfait 2002). Consequently, the numbers of species with flightless ability were highest in the older successional and more stable stages. Spider species are able to disperse by ballooning (Bell, Bohan, Shaw, & Weyman 2005), and as far as carabids are concerned, it can be expected that the inhabitants of dynamic grey dunes are mostly good and early colonizers (Bonte et al. 2002, 2006; Buchholz 2010). We found a contrasting pattern that might be related to larger body sizes in these habitats, since ballooning ability is associated with small species and juvenile stages (Bell et al. 2005). However it should be noted that the ballooning ability estimates were derived from the literature. This may have reduced the accuracy of our estimates, since ballooning propensity may vary among different populations and because literature data may refer to different communities and ecosystems. In (coastal) heathland ecosystems – often

consisting of a small-scale mosaic of different vegetation types – running can be a more adequate dispersal strategy as it enables fast and directed migration between grey dunes and nearby heathland or grass habitats, for example during unfavorable time periods (Bonte, Maelfait, & Hoffmann 2000; Bonte et al. 2002). Mass effects are also possible, as densely vegetated successional stages may constitute a source for grey dune spider assemblages, especially for lycosid species (Öberg, Ekbom, & Bommarco 2007).

Most carabids specialized in feeding on Collembola occurred in older successional stages, these being shady and more humid habitats that presumably favor higher Collembolan abundances. Active hunting spiders were dominant in open grey dunes, as previously shown by Finch (1997). Due to lower spatial resistance active hunting species, such as Lycosidae, are favored, while ambush and web hunters require well-developed habitat architecture, which is usually not given in grey dunes.

Conclusions

We present one of the first empirical studies to measure different components of functional diversity of ground-dwelling arthropods in response to coastal heathland succession. By means of the newly developed functional dispersion index (Laliberté & Legendre 2010) we showed that functional diversity of carabids and spiders differed along a successional gradient, while alpha-diversity remained unaffected. Functional diversity therefore appears to provide a useful tool to analyze various ways arthropods respond to environmental variability. Based on life-history trait shifts along the successional gradient we assume that trait-based environmental filtering plays a central role due to species sorting. This means that species sorting in which distinct life-history traits are favored at different successional stages is prevalent. Using the trait-based approach, including functional diversity indices such as FDis, could contribute valuable information for conservation. It is most important to maintain a mosaic of different successional stages, as functional diversity is enhanced at a landscape level and distinct life-history traits are promoted. Furthermore, conservation must focus on dynamic grey dunes, as these habitats filter (mostly rare and endangered) species adapted to extreme local conditions such as sand dynamics and hot microclimate.

Acknowledgements

We thank J. Mantilla-Contreras for help with the field work. Frank Dziok, Martin Entling, Klaus Hövemeyer and two anonymous reviewers gave helpful comments on an earlier version of the manuscript. We are grateful to the national park “Vorpommersche Boddenlandschaft” for permission to conduct the study in the protected area. The study was funded by the Bauer-Hollmann Foundation (Stifterverband für die

deutsche Wissenschaft) as part of the research project “Biodiversity, Ecology and Management of Coastal Habitats of the Baltic Sea”.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.baae.2012.08.015>.

References

- Baars, M. A. (1979). Catches in pitfall traps in relation to mean densities of Carabid beetles. *Oecologia*, 41, 25–46.
- Balmer, O., & Erhardt, A. (2000). Consequences of succession on extensively cultivated grasslands for Central European butterfly communities: Rethinking conservation practices. *Conservation Biology*, 14, 746–757.
- Bauer, L. (1972). *Handbuch der Naturschutzgebiete der Deutschen Demokratischen Republik. Band 1: Bezirke Rostock, Schwerin und Neubrandenburg*. Leipzig: Urania-Verlag.
- Bell, J. R., Bohan, D. A., Shaw, E. M., & Weyman, G. S. (2005). Ballooning dispersal using silk: World fauna, phylogenies, genetics and models. *Bulletin of Entomological Research*, 95, 69–114.
- Blake, S., Foster, G. N., Eyre, M. D., & Luff, M. L. (1994). Effects of habitat type and grassland management practice on the body size distribution of carabid beetles. *Pedobiologia*, 28, 502–512.
- Blaum, N., Seymour, C., Rossmanith, E., Schwager, M., & Jeltsch, F. (2009). Changes in arthropod diversity along a land use driven gradient of shrub cover in savanna rangelands: Identification of suitable indicators. *Biodiversity and Conservation*, 18, 1187–1199.
- Bonte, D., Baert, L., & Maelfait, J. P. (2002). Spider assemblage structure and stability in a heterogeneous coastal dune system (Belgium). *Journal of Arachnology*, 30, 331–343.
- Bonte, D., Lens, L., & Maelfait, J. P. (2006). Sand dynamics in coastal dune landscapes constrain diversity and life-history characteristics of spiders. *Journal of Applied Ecology*, 43, 735–747.
- Bonte, D., Maelfait, J. P., & Hoffmann, M. (2000). Seasonal and diurnal migration patterns of the spider (Araneae) fauna of coastal grey dunes. *Ekologia*, 19, 5–16.
- Bremner, J., Rogers, S. I., & Frid, C. L. (2006). Matching biological traits to environmental conditions in marine benthic ecosystems. *Journal of Marine Systems*, 60, 302–316.
- Brose, U. (2003). Bottom-up control of carabid beetle communities in early successional wetlands: Mediated by vegetation structure or plant diversity. *Oecologia*, 135, 407–413.
- Buchholz, S. (2010). Ground spider assemblages as indicators for habitat structure in inland sand ecosystems. *Biodiversity and Conservation*, 19, 2565–2595.
- Buddle, C. M., Spence, J. R., & Langor, D. W. (2000). Succession of boreal forest spider assemblages following wildfire and harvesting. *Ecography*, 23, 424–436.
- Cristofoli, S., Mahy, G., Kekenbosch, R., & Lambeets, K. (2010). Spider communities as evaluation tools for wet heathland restoration. *Ecological Indicators*, 10, 773–780.

- Da Silva, P. M., Aguiar, C. A. S., Niemelä, J., Sousa, J. P., & Serrano, A. R. M. (2007). Diversity patterns of ground-beetles (Coleoptera: Carabidae) along a gradient of land-use disturbance. *Agriculture, Ecosystems and Environment*, 124, 270–274.
- Diaz, A. M., Alonso, M. L. S., & Gutierrez, M. R. V. A. (2008). Biological traits of stream macroinvertebrates from a semi-arid catchment: Patterns along complex environmental gradients. *Freshwater Biology*, 53, 1–21.
- Entling, W., Schmidt-Entling, M. H., Bacher, S., Brandl, R., & Nentwig, W. (2010). Body size – Climate relationships of European spiders. *Journal of Biogeography*, 37, 233–237.
- Finch, O. D. (1997). Die Spinnen (Araneae) der Trockenrasen eines nordwestdeutschen Binnendünenkomplexes. *Drosera*, 97, 21–40.
- Folke, C. S., Carpenter, S., Walker, B., Scheffer, M., Elmquist, T., Gunderson, L., et al. (2004). Regime shifts, resilience, and biodiversity in ecosystem management. *Annual Review of Ecology, Evolution and Systematics*, 35, 557–581.
- Gajdos, P., & Toft, S. (2000). A twenty-year comparison of epigeic spider communities (Araneae) of Danish coastal heath habitats. *Journal of Arachnology*, 28, 90–96.
- Gerisch, M., Schanowski, A., Figura, W., Gerken, B., Dziok, F., & Henle, K. (2011). More species, but all do the same – Contrasting effects of ground beetle species and functional diversity to flood disturbance. *Oikos*, 121, 508–515.
- Gray, J. S. (1989). Effects of environmental stress on species rich assemblages. *Biological Journal of the Linnean Society*, 37, 19–32.
- Greenslade, P. J. M., & Southwood, T. R. E. (1962). The relationship of flight and habitat in some Carabidae (Coleoptera). *Entomologist*, 95, 86–88.
- Horvath, R., Magura, T., Szinetar, C., & Tothmeresz, B. (2009). Spiders are not less diverse in small and isolated grasslands, but less diverse in overgrazed grasslands: A field study (East Hungary, Nyirseg). *Agriculture, Ecosystems and Environment*, 130, 16–22.
- Huber, C., Schulze, C., & Baumgarten, M. (2007). The effect of femel- and small scale clear-cutting on ground dwelling spider communities in a Norway spruce forest in Southern Germany. *Biodiversity and Conservation*, 16, 3653–3680.
- Il Choi, W., Choi, K. S., Lyu, D. P., Lee, J. S., Lim, J., Lee, S., et al. (2010). Seasonal changes of functional groups in coleopteran communities in pine forests. *Biodiversity and Conservation*, 19, 2291–2305.
- Keil, P., Dziok, F., & Storch, D. (2008). Geographical patterns of hoverfly (Diptera, Syrphidae) functional groups in Europe: Inconsistency in environmental correlates and latitudinal trends. *Ecological Entomology*, 33, 748–757.
- Laliberté, E., & Legendre, P. (2010). A distance-based framework for measuring functional diversity from multiple traits. *Ecology*, 91, 299–305.
- Laliberté, E., & Shipley, B. (2010). *The FD package Version 1.0-9*. <http://cran.r-project.org/>
- Lambeets, K., Vandengehuchte, M. L., Maelfait, J. P., & Bonte, D. (2008). Understanding the impact of flooding on trait-displacements and shifts in assemblage structure of predatory arthropods on river banks. *Journal of Animal Ecology*, 77, 1162–1174.
- Lambeets, K., Vandengehuchte, M. L., Maelfait, J. P., & Bonte, D. (2009). Integrating environmental conditions and functional life-history traits for riparian arthropod conservation planning. *Biological Conservation*, 142, 625–637.
- Leibold, M. A., Holoyak, M., Mouquet, N., Amarasekare, P., Chase, J. M., Hoopes, M. F., et al. (2004). The metacommunity concept: A framework for multi-scale community ecology. *Ecology Letters*, 7, 601–613.
- Lin, Y. C., James, R., & Dolman, P. M. (2007). Conservation of heathland ground beetles (Coleoptera, Carabidae): The value of lowland coniferous plantations. *Biodiversity and Conservation*, 16, 1337–1358.
- Loreau, M. (2010). Linking biodiversity and ecosystems: Towards a unifying ecological theory. *Philosophical Transactions of the Royal Society B: Biological Science*, 365, 49–60.
- Lövei, G. L., Magura, T., Tothmeresz, B., & Ködöböcz, V. (2006). The influence of matrix and edges on species richness patterns of ground beetles (Coleoptera: Carabidae) in habitat islands. *Global Ecology and Biogeography*, 15, 283–289.
- Magurran, A. E. (2004). *Measuring biological diversity*. Malden, Oxford, Carlton: Blackwell Publishing.
- Mallis, R. E., & Hurd, L. E. (2005). Diversity among ground-dwelling spider assemblages: Habitat generalists and specialists. *Journal of Arachnology*, 33, 101–109.
- Marini, L., Fontana, P., Battisti, A., & Gaston, K. J. (2009). Response of orthopteran diversity to abandonment of semi-natural meadows. *Agriculture, Ecosystems and Environment*, 132, 232–236.
- Müller-Motzfeld, G. (Ed.). (2006). *Die Käfer Mitteleuropas. Band 2 Adephega 1: Carabidae (Laufkäfer)*. Heidelberg, Berlin: Spektrum Akademischer Verlag.
- Negro, M., Isaia, M., Palestini, C., Schoenhofer, A., & Rolando, A. (2010). The impact of high-altitude ski pistes on ground-dwelling arthropods in the Alps. *Biodiversity and Conservation*, 19, 1853–1870.
- Nentwig, W., Hänggi, A., Kropf, C., & Blick, T. (2003). *Central European Spiders. An internet identification key. Version of 08/12/2003*. <http://www.araneae.unibe.ch>
- New, T. R. (1998). *Invertebrate surveys for conservation*. Oxford: Oxford University Press.
- Niemelä, J., & Kotze, D. J. (2009). Carabid beetle assemblages along urban to rural gradients: A review. *Landscape and Urban Planning*, 92, 65–71.
- Öberg, S., Ekbom, B., & Bommarco, R. (2007). Influence of habitat type and surrounding landscape on spider diversity in Swedish agroecosystems. *Agriculture, Ecosystems and Environment*, 122, 211–219.
- Petchey, O. L., & Gaston, K. J. (2006). Functional diversity: Back to basics and looking forward. *Ecology Letters*, 9, 741–758.
- Petchey, O. L., Evans, K. L., Fishburn, I. S., & Gaston, K. J. (2007). Low functional diversity and no redundancy in British avian assemblages. *Journal of Animal Ecology*, 76, 977–985.
- Platnick, N. J. (2012). *The world spider catalog. Version 12.5*. <http://research.amnh.org/iz/spiders/catalog/index.html>
- Poff, N. L. (1997). Landscape filters and species traits: Towards mechanistic understanding and prediction in stream ecology. *Journal of the North American Benthological Society*, 16, 391–409.
- R Development Core Team. (2010). *R: A language and environment for statistical computing*. Vienna: R Foundation for Statistical Computing. www.R-project.org
- Reinhard, H. (1962). *Klimatologie. Atlas der Bezirke Rostock, Schwerin und Neubrandenburg*. Schwerin: VEB Topographischer Dienst.

- Remke, E. (2003). *Vegetationsökologischer Vergleich von Heidegebieten der Insel Hiddensee, Mecklenburg-Vorpommern*. Diploma thesis, University of Greifswald.
- Remmert, H. (1981). Body size of terrestrial arthropods and biomass of their populations in relation to the abiotic parameters of their milieu. *Oecologia*, 50, 12–13.
- Ribera, I., Dolédec, S., Downie, I. S., & Foster, G. N. (2001). Effect of land disturbance and stress on species traits of ground beetle assemblages. *Ecology*, 82, 1112–1129.
- Ribera, I., Foster, G. N., Downie, I. S., McCracken, D. I., & Abernethy, V. J. (1999). A comparative study of the morphology and life traits of Scottish ground beetles (Coleoptera, Carabidae). *Annales Zoologica Fennici*, 36, 21–37.
- Roberts, M. J. (1987). *The spiders of Great Britain and Ireland Volume 2: Linyphiidae and checklist*. Harley Books.
- Roberts, M. J. (1998). *Spinnen Gids*. Tirion.
- Ruzicka, V. (1985). The size groups in the spiders (Araneae) and carabids (Col, Carabidae). *Acta Universitatis Carolinae/Biologica*, 1982–1984, 77–107.
- Schirmel, J., & Buchholz, S. (2011). Response of carabid beetles (Coleoptera: Carabidae) and spiders (Araneae) to coastal heathland succession. *Biodiversity and Conservation*, 20, 1469–1482.
- Schirmel, J., Mantilla-Contreras, J., Blindow, I., & Fartmann, T. (2011). Impacts of succession and grass encroachment on heathland Orthoptera. *Journal of Insect Conservation*, 15, 633–642.
- Schirmel, J., Timler, L., & Buchholz, S. (2011). Impact of the invasive moss *Campylopus introflexus* on carabid beetles (Coleoptera: Carabidae) and spiders (Araneae) in acidic coastal dunes at the southern Baltic Sea. *Biological Invasions*, 13, 605–620.
- Schleuter, D., Daufresne, M., Massol, F., & Argillier, C. (2010). A user's guide to functional diversity indices. *Ecological Monographs*, 80, 469–484.
- Schultz, W., & Finch, O. D. (1996). *Biotoptypenbezogene Verteilung der Spinnenfauna der nordwestdeutschen Küstenregion – Charakterarten, typische Arten und Gefährdung*. Cuvillier Verlag.
- Tilman, D., Knops, J., Wedin, D., Reich, P., Ritchie, M., & Siemann, E. (1997). The influence of functional diversity and composition on ecosystem processes. *Science*, 277, 1300–1302.
- Turin, H. (2000). *De Nederlandse Loopkevers, verspreiding en oecologie (Coleoptera: Carabidae)*. Nederlandse fauna 3. Leiden: NNM Naturalis, KNNV Uitgeverij & EIS-Nederland.
- Umweltministerium Mecklenburg-Vorpommern. (2003). *Die Naturschutzgebiete in Mecklenburg-Vorpommern*. Schwerin: Demmler-Verlag.
- Vandewalle, M., de Bello, F., Berg, M. P., Bolger, T., Doledec, S., Dubs, F., et al. (2010). Functional traits as indicators of biodiversity response to land use changes across ecosystems and organisms. *Biodiversity and Conservation*, 19, 2921–2947.
- Villéger, S., Mason, N. W. H., & Mouillot, D. (2008). New multidimensional functional diversity indices for a multifaceted framework in functional ecology. *Ecology*, 89, 2290–2301.
- Violle, C., Navas, M. L., Vile, D., Kazakou, E., Fortunel, C., Hummel, I., et al. (2007). Let the concept of trait be functional!. *Oikos*, 116, 882–892.
- Webb, N. R. (1998). The traditional management of European heathlands. *Journal of Applied Ecology*, 35, 987–990.
- Zuur, A. F., Ieno, E. N., Walker, N. J., Saveliev, A. A., & Smith, G. M. (2009). *Mixed effects models and extensions in ecology with R*. New York: Springer.

Available online at www.sciencedirect.com

SciVerse ScienceDirect